Math 301 Proof Portfolio

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Introduction

Self Reflection

Throughout this semester I have learned and grown both as a mathematician and as a person. At the beginning of this semester, I struggled a lot with Math 301. The deadlines were suffocating, the content was baffling, and 9AM was far too early for me. But, as I worked through these challenges, I found that I could improve in all of these areas. It took some getting used to, but I worked hard and learned to meet the deadlines. Taking a look at my preparation assignments, at the beginning of the semester, I missed on average 1-2 preparation assignments a week. However, as the semester progressed, I became better at managing my time and stopped missing deadlines. Now, at the end of the semester, I have not missed a preparation assignment in over a month. As I learned to manage my time better, I stopped missing class as often. 9AM is early, but doable. One of the biggest improvements, however, was my skills in writing, reading, and thinking about math. Before I took this class I rarely thought deeply about math. In my classes, whenever I ran into a problem I didn't know how to do, I would either look it up or give up. I didn't bother to think about the why, try to figure things out, or use previous knowledge to help. As I have participated in Math 301, I have broken out of my fixed mindset about math. I find joy in thinking deeply about problems. My understanding of the basics of mathematics has grown significantly. I have noticeably grown as a person and a mathematician as a result of being a student in Math 301.

Grade Proposal

A-: I came to 80% of class periods and demonstrated a pattern of preparation throughout the semester by consistently posting my preparation on the discussion boards. I submitted 33 good faith weekly write-ups throughout the semester. I have received a complete on 16 polished proofs. I have at least one polished proof marked complete that demonstrates mastery for the first four topics/methods and 5 others (for a total of 9 topics/methods).

Topic/Method Mastery

This section lists the 10 topics/methods that are central ideas in Math 301, along with the Problem number/name where I demonstrate that topic/method.

1. proving two sets are equal
    - Problem 8 - First Proof That Two Sets Are Equal
    - Problem 10 - Intersection Of Two Intervals
2. proving statements involving universal and existential quantifiers
    - Problem 24 - Practice Finding Truth Values With Universal Quantifiers
3. proving that a sequence converges to a specific number
    - Problem 41 - Showing A Sequence Converges
    - Problem 57 - Proving A Quotient Of Two Quadratic Sequences Converges
4. proving a statement is true by induction
    - Problem 33 - First Induction Problem
5. proving a number is an infimum or supremum of a set
    - Problem 3.1 - Practice With Definition Of Infimum And Supremum
    - Problem 5 - Using The Completeness Axiom
6. proving a number is a limit point of a set
    - Problem 6.1 - A Limit Point Of An Open Interval
    - Problem 7 - A Set With One Limit Point
    - Problem 27 - Limit Points Of Subsets Are Limit Points Of The Larger Set
7. proving something is a function, an injective function, and/or a surjective function
    - Problem 35.3 - Practice With Bijective
    - Problem 54 - The Composition Of Injective Functions Is Injective
8. proving a statement is true using an indirect proof (proof by contrapositive and/or contradiction)
    - Problem 5 - Using The Completeness Axiom
    - Problem 8 - First Proof That Two Sets Are Equal
    - Problem 24 - Practice Finding Truth Values With Universal Quantifiers
9. proving facts about images and preimages of functions
    - Problem 44 - Image And Preimage Property 3
10. proving properties about convergent sequences in general
    - not done

All proofs

Problem (3.1)**.** Let $T = [-1, 2)$. Prove that -1 is the greatest lower bound of T. In other words, show that $\inf T = -1$. Then prove that $\sup T = 2$

Solution.
We will show that $\inf T = -1$ for $T = [-1, 2)$. Let $x \in T$. Therefore, we know $-1 \leq x$. Repeat for every x. Since -1 is less than or equal to all elements of T, we know that -1 is a lower bound of T. Let m be a lower bound of T. Because m is a lower bound of T, we know m is less than or equal to all elements of T. We know that $-1 \geq -1$ and $-1 < 2$, therefore $-1 \in T$. Because $-1 \in T$ and m is a lower bound of T, we know that $m \leq -1$. Repeat for every m. Since -1 is greater than or equal to all lower bounds of T, and -1 is a lower bound of T, we can see that -1 is the infimum of T, or $\inf T = -1$.

We will show that $\sup T = 2$ for $T = [-1, 2)$. Let $y \in T$. Therefore, we know $y < 2$, so $2 \geq y$. Repeat for every y. Since 2 is greater than all elements of T, we know that 2 is an upper bound of T.

Let k be an upper bound of T. Because k is an upper bound of T, we know that it is greater than or equal to all elements of T. Suppose by way of contradiction that $k < 2$. Because k and 2 are two distinct real numbers, we know that there exists a real number between them. Let q be a real number between k and 2 such that $k < q < 2$. Let $x \in T$. Because $x \in T$ we know $-1 \leq x < 2$. Because $x \in T$ and k is an upper bound of T we know that $k \geq x$. We know that $k \geq x$ and $x \geq -1$, therefore $k \geq -1$. Because $k \geq -1$ and $q > k$, we see that $q > -1$. Therefore, $q \geq -1$. Since $q \geq -1$ and $q < 2$, we know that $q \in T$. Because k is an upper bound of T and $q \in T$ we know that $k \geq q$. This is a contradiction because k cannot be both less than and greater than or equal to q. Therefore, we know that $k \geq 2$. Repeat for every k. Since 2 is less than or equal to all upper bounds of T, and 2 is an upper bound of T, we can see that 2 is the supremum of T, or $\sup T = 2$.

Problem (5)**.** Consider the solutions to the inequality $x^2 < 2$. We can write this using set builder notation as $S = \{x \mid x^2 < 2\}$. Identify $\sup S$, or explain why there is no supremum of S.

Solution.
We will prove that $\sup S = \sqrt{2}$ for $S = \{x \mid x^2 < 2\}$. Let $x \in S$. Therefore, we know that $x^2 < 2$, so $x < \sqrt{2}$ and $x > -\sqrt{2}$. Because $x < \sqrt{2}$ we know $\sqrt{2} \geq x$. Repeat for every x. Since $\sqrt{2}$ is greater than or equal to all elements in S, we see that $\sqrt{2}$ is an upper bound of S.

Let m be an upper bound of S. Suppose by way of contradiction that $m < \sqrt{2}$. We know that m and $\sqrt{2}$ are two distinct real numbers, therefore there exists a real number between them. Let z be a real number between m and $\sqrt{2}$, such that $m < z < \sqrt{2}$. Because $1^2 = 1$ and $1 < 2$ we see $1 \in S$. Because $1 \in S$ and $1 > 0$ we see that there exists a number in the set S that is positive. Because there exists a number in S that is positive and m is an upper bound of S, we know that $m > 0$, or m is positive. Since $m > 0$ and $m < z$, we know that $z > 0$. Since $0 < z < \sqrt{2}$, we know that $z^2 < 2$. Therefore, we know $z \in S$. Because $z \in S$ and m is an upper bound of S, we see that $m \geq z$. This is a contradiction, as m cannot be less than and greater than or equal to z. Therefore, we know $m \geq \sqrt{2}$. Repeat for every m.

Since $\sqrt{2}$ is an upper bound of S and for every upper bound m we know $\sqrt{2} \leq m$, we know that $\sqrt{2}$ must be the supremum of S. Therefore, we know $\sup S = \sqrt{2}$.

Problem (6.1)**.** Let $S = (0, 1)$ which is the open interval from 0 to 1 that does not include the end points. Prove that 1 is a limit point of S.

Solution.
Let $S = (0, 1)$. We will prove that 1 is a limit point of S. Let I be an open interval (a, b) such that $a < 1 < b$. Suppose $a \geq 0$. We know that $a \neq 1$, therefore there must be a number x such that $a < x < 1$. Let x be that value. Since $0 \leq a$ and $a < x$ we know $0 < x$. Since $x > 0$ and $x < 1$ we know $x \in S$. Since $x < 1 < b$ and $x > a$, we know $x \in I$. Because $x < 1$, we know $x \neq 1$.

Now suppose $a < 0$. Let $y \in S$. Since $y \in S$, we know $0 < y < 1$. Because $a < 0$ and $0 < y$ we see $a < y$. Because $y < 1$ and $1 < b$ we see $y < b$. Because $a < y$ and $y < b$, we see that $y \in I$. Because $y < 1$, we know $y \neq 1$. Repeat for every I. Since every interval that includes 1 includes another number in S that is not equal to 1, we know that 1 is a limit point of S.

Problem (7)**.** Let $S = \left\{\frac{1}{n} \mid n \in \mathbb{N}\right\}$, the collection of fractions of the form $\frac{1}{n}$ where n is a natural number. Prove that 0 is a limit point of S.

Solution.
We will show that 0 is a limit point of the set $S = \left\{\frac{1}{n} \mid n \in \mathbb{N}\right\}$. Let I be an open interval (a, b) that includes 0. We know $a < 0 < b$. We will consider two cases. First, where $b > 1$ and second, where $b \leq 1$.

Suppose $b > 1$. Let $\frac{1}{y} \in S$. Because $\frac{1}{y} \in S$, we know that y is a natural number, therefore $y > 0$. We know $1 > 0$. We know that when you divide a positive number by a positive number, you get a positive number, therefore $\frac{1}{y} > 0$. Because y is a natural number, we also know that $y \geq 1$. For a real number x, if $x \geq 1$, we know $\frac{1}{x} \leq 1$. We know $y \geq 1$, thus we see that $\frac{1}{y} \leq 1$. We know $\frac{1}{y} \leq 1$ and $b > 1$, therefore we know $\frac{1}{y} \leq 1 < b$, or $\frac{1}{y} < b$. We know $\frac{1}{y} > 0$ and $a < 0$, therefore we know $a < 0 < \frac{1}{y}$ or $a < \frac{1}{y}$. Since $\frac{1}{y} < b$ and $a < \frac{1}{y}$, we see $\frac{1}{y} \in I$. Since $\frac{1}{y} > 0$, we know $\frac{1}{y} \neq 0$.

Now suppose $b \leq 1$. Let x be a natural number where $x > \frac{1}{b}$. Because x is a natural number, we know $\frac{1}{x} \in S$ and $x > 0$. Since we know $b > 0$, $x > 0$, and $x > \frac{1}{b}$, we know $b > \frac{1}{x}$. Thus we see that $b > \frac{1}{x}$. Because $x > 0$ and $1 > 0$, we know $\frac{1}{x} > 0$. Since $\frac{1}{x} > 0$ and $0 > a$, we know $\frac{1}{x} > a$. Because $a < \frac{1}{x} < b$, we know $\frac{1}{x} \in I$. Since $\frac{1}{x} > 0$, we know $\frac{1}{x} \neq 0$.

Repeat for every I. Thus we see that every interval that contains 0 also contains an element of S that is not 0. Therefore, we know 0 is a limit point of S.

Problem (8)**.** Let $I = (5, 9]$. Consider the sets $S = \{x \mid x \text{ is an upper bound of } I\}$ and $A = \{x \mid x \geq 9\}$. Prove that $S = A$.

Solution.
Let $I = (5, 9]$. Let $S = \{x \mid x \text{ is an upper bound of } I\}$. Let $A = \{x \mid x \geq 9\}$. We will prove that $S = A$. Let $x \in S$. We know that x is an upper bound of I. Therefore, we know x is greater than or equal to every element of I. Suppose by way of contradiction that $x < 9$. Since x and 9 are two distinct real numbers, we know that there exists a real number such that $x < n < 9$. Let n be that number. We know $n < 9$, therefore we see that $n \leq 9$. We will consider two cases, one where $n > 5$ and the other where $n \leq 5$. Suppose $n > 5$. Because $5 < n \leq 9$, we know that $n \in I$. Because x is an upper bound of I and $n \in I$, we see that $x \geq n$. This is a contradiction, as $x < n$ and $x \geq n$ cannot both be true. Now suppose that $n \leq 5$. Let $y \in I$. We know $5 < y \leq 9$. Therefore, we see that $n \leq 5 < y$. Since $n < y$ and $x < n$, we see that $x < y$. Because x is an upper bound of I and $y \in I$, we see that $x \geq y$. This is a contradiction, as $x < y$ and $x \geq y$ cannot both be true. Therefore, in this case, we know $x \geq 9$. Thus we see that $x \geq 9$. Since in both cases we reached a contradiction, we know that $x \geq 9$. Because $x \geq 9$, we know $x \in A$. Repeat for every x. Since every element of S is an element of A, we know $S \subseteq A$.

Let $z \in A$. Let $m \in I$. We know that $m \leq 9$ and that $9 \leq z$, therefore $m \leq z$. Repeat for every m. Thus we see that z is greater than or equal to every element of I. Therefore, we know $z \in S$. Repeat for every z. Thus we see that every element of A is also an element of S. Since every element of A is an element of S, we know $A \subseteq S$. Since we know $S \subseteq A$ and $A \subseteq S$, we see that $S = A$.

Problem (10)**.** Suppose that $a, b, c, d \in \mathbb{R}$ and that $a < b < c < d$. Consider the intervals $A = (a, c)$ and $B = [b, d]$. Prove that $A \cap B = [b, c)$.

Solution.
Let $A = (a, c)$ and $B = [b, d]$. We will show that $A \cap B = [b, c)$. Let $U = [b, c)$. Let $V = A \cap B$. Let $x \in V$. Therefore, we know that $x \in A$ and $x \in B$. Because $x \in A$ we know $x < c$. Since $x \in B$ we see $b \leq x$. Since we know $x < c$ and $x \geq b$, we see that $b \leq x < c$. Therefore, $x \in U$. Repeat for every x. Thus we see that every element of V is also an element of U.

Let $y \in U$. We know $b \leq y < c$. Since $a < b < c < d$ and $b \leq y < c$, we know $a < b \leq y < c < d$. Therefore, we know $a < y < c$ and $b \leq y \leq d$. Since $a < y < c$, we know $y \in A$. Since $b \leq y \leq d$, we know $y \in B$. Because $y \in A$ and $y \in B$, we see that $y \in V$. Repeat for every y. Thus we see that every element of U is also an element of V.

Since every element of U is an element of V and every element of V is an element of U, we conclude that $V = U$. Therefore, we know $A \cap B = [b, c)$.

Problem (24)**.** Determine the truth value of each statement below. Justify your claims.

1. $\forall x \in \mathbb{R}$ and $\forall y \in \mathbb{R}$, $\exists z \in \mathbb{R}$ such that $x + y = z$.
2. $\forall x \in \mathbb{R}$ and $\forall y \in \mathbb{R}$, $\exists z \in \mathbb{R}$ such that $xz = y$.
3. $\forall x \in \mathbb{R}$, $\exists y \in \mathbb{R}$ such that $\forall z \in \mathbb{R}$, $z > y$ implies $z > x + y$.
4. $\exists x \in \mathbb{R}$ such that $\forall y \in \mathbb{R}$, $\exists z \in \mathbb{R}$ such that $z > y$ implies $z > x + y$.

Solution.
Statement 1: $\forall x \in \mathbb{R}$ and $\forall y \in \mathbb{R}$, $\exists z \in \mathbb{R}$ such that $x + y = z$.

We will show that this statement is true. Let $x \in \mathbb{R}$. Let $y \in \mathbb{R}$. We know that when we add two real numbers the result is another real number. Therefore, we know that when x and y are added, the result is another real number. Let $z = x + y$. Repeat for every x and y. Thus we see that for every real number x and y there exists a real number z where $x + y = z$.

Statement 2: $\forall x \in \mathbb{R}$ and $\forall y \in \mathbb{R}$, $\exists z \in \mathbb{R}$ such that $xz = y$.

We will show that this statement is false by proving that its negation is true. We will show that there exists $x \in \mathbb{R}$ and $y \in \mathbb{R}$ such that for every $z \in \mathbb{R}$, we know $xz \neq y$ is true. Let $x = 0$. Let $y = 1$. Let $z \in \mathbb{R}$. We know that when you multiply any number by 0, it will equal 0. Therefore, $(0)z = 0$. Since $x = 0$ and $(0)z = 0$ we see that $xz = 0$. We know that $0 \neq 1$, therefore $xz \neq y$. Repeat for every z. We see that there exists a real number x and y such that $xz \neq y$ is true for every real number z. Because the negation is true, we see that the original statement is false.

Statement 3: $\forall x \in \mathbb{R}$, $\exists y \in \mathbb{R}$ such that $\forall z \in \mathbb{R}$, $z > y$ implies $z > x + y$.

We will show that this statement is false by proving that its negation is true. We will show that there exists $x \in \mathbb{R}$ such that for every $y \in \mathbb{R}$ there exists $z \in \mathbb{R}$ such that $z > y$ and $z \leq x + y$ is true. Let $x = 100$. Let $y \in \mathbb{R}$. Let $z = y + \frac{x}{2}$. We compute

$$\begin{aligned} z &= y + \frac{x}{2} && \text{(replace using assumption)} \\ &= y + \frac{100}{2} && \text{(replace using assumption)} \\ &= y + 50 && \text{(simplify).} \end{aligned}$$

We see that $z = y+50$, therefore $z = y+50 > y$ or $z > y$. We know that $x+y = y+100$ and $y+50 \leq y+100$, therefore, we see that $z \leq x + y$. Repeat for all y. We see that there exists a real number x such that for every real number y, there exists a real number z such that $z > y$ and $z \leq x + y$. Because the negation is true, we see that the original statement is false.

Statement 4: $\exists x \in \mathbb{R}$ such that $\forall y \in \mathbb{R}$, $\exists z \in \mathbb{R}$ such that $z > y$ implies $z > x + y$.

We will show that this statement is true. Let $x = 9$. Let $y \in \mathbb{R}$. Let $z = x + y + 1$. We compute

$$\begin{aligned} z &= x + y + 1 && \text{(replace using assumption)} \\ &= 9 + y + 1 && \text{(replace using assumption)} \\ &= y + 10 && \text{(simplify).} \end{aligned}$$

We see that $z = y + 10$, therefore $z = y + 10 > y$ or $z > y$. We know that $x + y = y + 9$ and $y + 10 > y + 9$, therefore $z > x + y$. Repeat for all y. We see that there exists a real number x such that for every real number y, there exists a real number z such that when $z > y$, we know $z > x + y$.

Problem (27)**.** Suppose that A and B are subsets of the real numbers. Prove that if $A \subseteq B$ and p is a limit point of A, then p is a limit point of B.

Solution.
Let A and B be subsets of the real numbers. We will prove that if $A \subseteq B$ and p is a limit point of A, then p is a limit point of B. Suppose that $A \subseteq B$. Let p be a limit point of A. Let I be an open interval where I includes p. Because p is a limit point of A, we know that there exists a number in I that is also in A that is not equal to p. Let x be such a number. We know $x \in A$, $x \in I$, and $x \neq p$. Since $A \subseteq B$ and $x \in A$, we see that $x \in B$. Repeat for every I. Because $x \in I$, $x \in B$, and $x \neq p$, we see that every interval that contains p also contains a point that is not equal to p that is in B. Therefore, p is a limit point of B.

Problem (33)**.** Notice that

$$\begin{aligned} 1 &= 1 \\ 1+3 &= 4 \\ 1+3+5 &= 9 \\ 1+3+5+7 &= 16 \\ 1+3+5+7+9 &= 25. \end{aligned}$$

Make a conjecture about the sum of odd numbers by giving a general formula for what you see above. Then prove that your formula is valid by using mathematical induction.

Solution.
We will prove that, for all natural numbers x, we know the statement $(2(1)-1)+(2(2)-1)+...+(2x-1) = x^2$ is true. If we simplify the expression $(2(1)-1)+(2(2)-1)+...+(2x-1)$ we find that it is equal to $1+3+5+...+(2x-1)$. Let $S = \{x \in \mathbb{N} \mid 1+3+5+...+(2x-1) = x^2\}$. We will show that S is the set of all natural numbers, or $S = \mathbb{N}$. We will use the Principle of Mathematical Induction. We know that $2(1)-1 = 1 = 1^2$, therefore $1 \in S$. Let $k \in \mathbb{N}$. Suppose $k \in S$. We know $1+3+5+...+(2k-1) = k^2$. We want to show $1+3+5+...+(2k-1)+(2(k+1)-1) = (k+1)^2$. We compute:

$$\begin{aligned} \underbrace{1+2+3+\cdots+(2k-1)}+(2(k+1)-1) &= k^2+(2(k+1)-1) && \text{(replace using the assumption)} \\ &= k^2+2k+2-1 && \text{(simplify)} \\ &= k^2+2k+1 && \text{(simplify)} \\ &= (k+1)^2 && \text{(factor).} \end{aligned}$$

Thus we see that $1+3+5+...+(2k-1)+(2(k+1)-1) = (k+1)^2$. Therefore, we know $k+1 \in S$. Because we know $k+1 \in S$, we see that if an element $k \in S$ then $k+1 \in S$. Since we know $1 \in S$ and if an element $k \in S$ then $k+1 \in S$ we know by the Principle of Mathematical Induction that $S = \mathbb{N}$.

Problem (35.3)**.** Let $h : A \to B$ be defined by $h(x) = x^2 + 4x + 4$. Select intervals A and B so that h is bijective. Prove your claim.

Solution.
Let $A = [-2, \infty)$. Let $B = [0, \infty)$. Let $h : A \to B$ be defined by $h(x) = x^2 + 4x + 4$. Let $a, b \in A$. Because $A = [-2, \infty)$, we know $a \geq -2$ and $b \geq -2$. Suppose $h(a) = h(b)$. We know $h(a) = a^2 + 4a + 4 = (a + 2)^2$, or $h(a) = (a + 2)^2$. We know $h(b) = b^2 + 4b + 4 = (b + 2)^2$, or $h(b) = (b + 2)^2$. Because $h(a) = (a + 2)^2$, $h(b) = (b+2)^2$, and $h(a) = h(b)$, we see that $(a+2)^2 = (b+2)^2$. Therefore, we know $\sqrt{(a + 2)^2} = \sqrt{(b + 2)^2}$. We know that $(a + 2)^2 \geq 0$. Hence we see $\sqrt{(a + 2)^2} = (a + 2)$. We know that $(b + 2)^2 \geq 0$. Hence we see $\sqrt{(b + 2)^2} = b + 2$. Because $\sqrt{(a + 2)^2} = a + 2$, $\sqrt{(b + 2)^2} = b + 2$, and $\sqrt{(a + 2)^2} = \sqrt{(b + 2)^2}$, we know that $a + 2 = b + 2$. Therefore, we see $a = b$. Repeat for every $a, b \in A$. Thus we see that for every $a, b \in A$, if $h(a) = h(b)$ then $a = b$. Therefore, we know h is injective. Let $y \in B$. Because $B = [0, \infty)$, we know $y \geq 0$. Therefore, we see $\sqrt{y} \in \mathbb{R}$. Let $x = \sqrt{y} - 2$. We compute

$$\begin{aligned}
h(x) &= x^2 + 4x + 4 && \text{(replace using assumption)} \\
&= (\sqrt{y} - 2)^2 + 4(\sqrt{y} - 2) + 4 && \text{(replace using assumption)} \\
&= y - 4\sqrt{y} + 4 + 4\sqrt{y} - 8 + 4 && \text{(multiply)} \\
&= y && \text{(simplify).}
\end{aligned}$$

Thus we see that $h(x) = y$. Repeat for every $y \in B$. Thus we see that for every $y \in B$ there exists an x such that $h(x) = y$. Therefore, we know h is surjective. Because h is both injective and surjective, we know h is bijective.

Problem (41)**.** Consider the sequence $(a_n) = \left(\frac{n+1}{n}\right)$. Prove that (a_n) converges to $L = 1$.

Solution.
Let $(a_n) = (\frac{n+1}{n})$. Let $\varepsilon > 0$. Let $M = \frac{1}{\varepsilon}$. Let $n \in \mathbb{N}$. We know $a_n = \frac{n+1}{n}$. Because $n \in \mathbb{N}$, we know $n > 0$. Therefore, we know $\frac{1}{n} > 0$. We compute

$$\begin{aligned}
|a_n - 1| &= \left|\frac{n+1}{n} - 1\right| && \text{(replace with assumption)} \\
&= \left|\frac{n+1}{n} - \frac{n}{n}\right| && \text{(find a common denominator)} \\
&= \left|\frac{n+1-n}{n}\right| && \text{(subtract)} \\
&= \left|\frac{1}{n}\right| && \text{(simplify)} \\
&= \frac{1}{n} && (\frac{1}{n} > 0).
\end{aligned}$$

Thus we see that $|a_n - 1| = \frac{1}{n}$. Suppose $n > M$. We want to show that $\varepsilon > |a_n - 1|$. We know $n > M$ and $M = \frac{1}{\varepsilon}$, therefore we see $n > \frac{1}{\varepsilon}$. Because $\varepsilon > 0$ and $n > \frac{1}{\varepsilon}$, we know $n\varepsilon > 1$. Because $n > 0$ and $n\varepsilon > 1$, we see $\varepsilon > \frac{1}{n}$. Because $\varepsilon > \frac{1}{n}$ and $\frac{1}{n} = |a_n - 1|$, we know $\varepsilon > |a_n - 1|$. Repeat for every n. Thus we see that for every $n \in \mathbb{N}$, when $n > M$ we know $\varepsilon > |a_n - 1|$. Repeat for every ε. Thus we see that for every $\varepsilon > 0$, there exists a real number M such that for every natural number n, when $n > M$ we know $\varepsilon > |a_n - 1|$. Therefore, we know that (a_n) converges to 1.

Problem (44)**.** Prove property 3 for images. So let $f : X \to Y$. Then prove that if $A_1 \subseteq X$ and $A_2 \subseteq X$, then we have $f(A_1 \cap A_2) \subseteq f(A_1) \cap f(A_2)$.

Solution.
Let $f : X \to Y$. Let $A \subseteq X$. Let $B \subseteq X$. We will show that if $A \subseteq X$ and $B \subseteq X$, then we have $f(A \cap B) \subseteq f(A) \cap f(B)$. Because $A \subseteq X$ and $B \subseteq X$, we know $A \cap B \subseteq X$. Therefore, we know that $f(A \cap B)$ exists. We will consider two cases. The first is where $A \cap B = \emptyset$ and the second is where $A \cap B \neq \emptyset$. Suppose $A \cap B = \emptyset$. Because $A \cap B = \emptyset$, we see that $f(A \cap B) = \emptyset$. We know that the empty set is a subset of every set. Therefore, we know $\emptyset \subseteq f(A) \cap f(B)$. Because $f(A \cap B) = \emptyset$ and $\emptyset \subseteq f(A) \cap f(B)$, we see that $f(A \cap B) \subseteq f(A) \cap f(B)$.

Now suppose $A \cap B \neq \emptyset$. Because $A \cap B \neq \emptyset$, we see that $f(A \cap B) \neq \emptyset$. Let $x \in f(A \cap B)$. Therefore, we know there exists a $y \in A \cap B$, such that $f(y) = x$. Let y be that number. Therefore, we know $y \in A \cap B$ and $f(y) = x$. Because $y \in A \cap B$, we see that $y \in A$ and $y \in B$. Since $y \in A$, we know $f(y) \in f(A)$. Because $y \in B$, we see that $f(y) \in f(B)$. Because $f(y) \in f(A)$ and $f(y) = x$, we know $x \in f(A)$. Since $f(y) \in f(B)$ and $f(y) = x$, we see that $x \in f(B)$. Because $x \in f(A)$ and $x \in f(B)$, we see that $x \in f(A) \cap f(B)$. Repeat for every x. Thus we see that every element of $f(A \cap B)$ is an element of $f(A) \cap f(B)$. Therefore, we know $f(A \cap B) \subseteq f(A) \cap f(B)$.

Problem (49)**.** Let A, B, and C be sets. Prove that

1. $A \cup (B \cup C) = (A \cup B) \cup C$.
2. $A \cap (B \cap C) = (A \cap B) \cap C$.

Solution.

1. Let $w \in A \cup (B \cup C)$. Therefore, we know that $w \in A$ or $w \in (B \cup C)$. Suppose $w \in A$. Because $w \in A$, we know that $w \in (A \cup B)$. Therefore, we see that $w \in (A \cup B) \cup C$. Now, suppose $w \in (B \cup C)$. Hence, we see $w \in B$ or $w \in C$. Suppose $w \in B$. Because $w \in B$, we know that $w \in (A \cup B)$. Therefore, we know $w \in (A \cup B) \cup C$. Now, suppose $w \in C$. Since $w \in C$, we know $w \in (A \cup B) \cup C$. Repeat on w. Thus we see that every element of $A \cup (B \cup C)$ is also an element of $(A \cup B) \cup C$. Therefore, we see $A \cup (B \cup C) \subseteq (A \cup B) \cup C$. Let $x \in (A \cup B) \cup C$. Therefore, we know that $x \in C$ or $x \in (A \cup B)$. Suppose $x \in C$. Because $x \in C$, we know that $x \in (B \cup C)$. Therefore, we see that $x \in A \cup (B \cup C)$. Now, suppose $x \in (A \cup B)$. Hence, we see $x \in A$ or $x \in B$. Suppose $x \in A$. Because $x \in A$, we know that $x \in A \cup (B \cup C)$. Now, suppose $x \in B$. Since $x \in B$, we know $x \in (B \cup C)$. Therefore, we see $x \in A \cup (B \cup C)$. Repeat on x. Thus we see that every element of $(A \cup B) \cup C$ is also an element of $A \cup (B \cup C)$. Therefore, we see $(A \cup B) \cup C \subseteq A \cup (B \cup C)$. Because $A \cup (B \cup C) \subseteq (A \cup B) \cup C$ and $(A \cup B) \cup C \subseteq A \cup (B \cup C)$, we know $A \cup (B \cup C) = (A \cup B) \cup C$.
2. Let $y \in A \cap (B \cap C)$. Because $y \in A \cap (B \cap C)$, we know that $y \in A$ and $y \in (B \cap C)$. Because $y \in (B \cap C)$, we know that $y \in B$ and $y \in C$. Because $y \in A$ and $y \in B$, we see that $y \in (A \cap B)$. Because $y \in (A \cap B)$ and $y \in C$, we know $y \in (A \cap B) \cap C$. Repeat on y. Thus we see that every element of $A \cap (B \cap C)$ is also an element of $(A \cap B) \cap C$. Therefore, we know $A \cap (B \cap C) \subseteq (A \cap B) \cap C$. Let $z \in (A \cap B) \cap C$. Because $z \in (A \cap B) \cap C$, we know that $z \in (A \cap B)$ and $z \in C$. Because $z \in (A \cap B)$, we know that $z \in A$ and $z \in B$. Because $z \in B$ and $z \in C$, we see that $z \in (B \cap C)$. Because $z \in A$ and $z \in (B \cap C)$, we know $z \in A \cap (B \cap C)$. Repeat on z. Thus we see that every element of $(A \cap B) \cap C$ is also an element of $A \cap (B \cap C)$. Therefore, we know $(A \cap B) \cap C \subseteq A \cap (B \cap C)$. Because $A \cap (B \cap C) \subseteq (A \cap B) \cap C$ and $(A \cap B) \cap C \subseteq A \cap (B \cap C)$, we know $A \cap (B \cap C) = (A \cap B) \cap C$.

Problem (54)**.** Let A, B, and C be sets, and consider the functions $f : A \to B$ and $g : B \to C$. Prove that if both f and g are injective, then $g \circ f$ is injective.

Solution.
Let $f : A \to B$ and $g : B \to C$. Thus we see that $g \circ f : A \to C$. We will show that if g and f are injective, then $g \circ f$ is injective. Suppose f and g are injective. Let $a, b \in A$. Because $a, b \in A$ and $g \circ f : A \to C$, we know $(g \circ f)(a)$ and $(g \circ f)(b)$ exist. We know that $(g \circ f)(a) = g(f(a))$ and $(g \circ f)(b) = g(f(b))$. Suppose $(g \circ f)(a) = (g \circ f)(b)$. Since $(g \circ f)(a) = (g \circ f)(b)$, $(g \circ f)(a) = g(f(a))$, and $(g \circ f)(b) = g(f(b))$ we see that $g(f(a)) = g(f(b))$. Because g is injective and $g(f(a)) = g(f(b))$, we know $f(a) = f(b)$. Since f is injective and $f(a) = f(b)$, we see that $a = b$. Repeat for every a and b. Thus we see that if $(g \circ f)(a) = (g \circ f)(b)$, we know $a = b$. Therefore, we know $g \circ f$ is injective. Thus we see that if g and f are injective, we know $g \circ f$ is injective.

Problem (57)**.** Prove that $\left(\frac{n^2}{n^2+1}\right)$ converges.

Solution.
Let $(a_n) = \left(\frac{n^2}{n^2+1}\right)$. Let $\varepsilon > 0$. Let $M = \frac{1}{\varepsilon}$. Let $n \in \mathbb{N}$. Suppose $n > M$. Since $\varepsilon > 0$, we see that $\frac{1}{\varepsilon} > 0$. Because $\frac{1}{\varepsilon} > 0$ and $M = \frac{1}{\varepsilon}$, we know $M > 0$. Since $n \in \mathbb{N}$, we know $n \geq 1$. Therefore, we see that $n^2 \geq n$. We want to show that $|a_n - 1| < \varepsilon$. We compute:

$$\begin{aligned}
|a_n - 1| &= \left|\frac{n^2}{n^2+1} - 1\right| && \text{(replace using assumption)} \\
&= \left|\frac{n^2}{n^2+1} - \frac{n^2+1}{n^2+1}\right| && \text{(simplify)} \\
&= \left|\frac{n^2 - n^2 - 1}{n^2+1}\right| && \text{„} \\
&= \left|\frac{-1}{n^2+1}\right| && \text{„} \\
&= \frac{1}{n^2+1} && (-1 < 0 \text{ and } n^2 + 1 > 0) \\
&< \frac{1}{n^2} && (n^2 + 1 > n^2) \\
&\leq \frac{1}{n} && (n^2 \geq n) \\
&< \frac{1}{M} && (n > M) \\
&= \frac{1}{\frac{1}{\varepsilon}} && \text{(replace using assumption)} \\
&= \varepsilon && \text{(simplify).}
\end{aligned}$$

Thus we see that $|a_n - 1| < \varepsilon$. Repeat on n. Repeat on ε. Thus we see that (a_n) converges to 1.

Problem (64)**.** Suppose that S and T are nonempty bounded subsets of $\mathbb{R}$ and that $S \subseteq T$. Prove that $\inf T \leq \inf S \leq \sup S \leq \sup T$.

Solution.
Let S, T be nonempty, bounded subsets of $\mathbb{R}$. Suppose $S \subseteq T$. Because S, T are nonempty, bounded subsets of $\mathbb{R}$, we know that $\inf T$, $\sup T$, $\inf S$, and $\sup S$ exist. Let $a = \inf T$, $b = \inf S$, $c = \sup S$, and $d = \sup T$. Let $x \in S$. Because $S \subseteq T$ and $x \in S$, we know that $x \in T$. Because $x \in T$ and $a = \inf T$, we know $a \leq x$. Repeat for every x. Thus we see that a is less than or equal to every element of S. Therefore, a is a lower bound of S. Because a is a lower bound of S and $b = \inf S$, we see that $a \leq b$. Because $x \in T$ and $d = \sup T$, we know $d \geq x$. Repeat for every x. Thus we see that x is greater than or equal to every element of S. Therefore, d is an upper bound of S. Because d is an upper bound of S and $c = \sup S$, we see that $c \leq d$. Because $x \in S$ and $b = \inf S$, we know that $b \leq x$. Because $x \in S$ and $c = \sup S$, we know that $x \leq c$. Since we know $b \leq x$ and $x \leq c$, we see that $b \leq c$. Because $a \leq b$, $b \leq c$, and $c \leq d$, we know $a \leq b \leq c \leq d$. Therefore, we see $\inf T \leq \inf S \leq \sup S \leq \sup T$.